

CHAPTER 10

INTRODUCTION TO STATISTICAL INFERENCE

10.1 OVERVIEW

The first 10 chapters of the book were devoted to probabilistic foundations of the theory of statistics, we introduced and developed techniques useful for predicting the form of future observations (data). Given certain general information (e.g., about independence of some events), we could deduce the distributions of observed random variables and answer questions such as: How many future observations will fall into a certain set? What will be the average of those observations?

In statistics (more precisely, in inferential statistics), which will be now our main object of study, the question is: Given the data, what can we say about specific aspects of the stochastic mechanisms that govern the occurrence of those data? The actual data are regarded as a result of a random process, in the sense that if the data collection were to be repeated, the outcome would most likely be different, possibly even leading to a different conclusion. Consequently, whatever inference we make from the actual data, it is subject to error. This error—the central concept of statistics—is not meant to be a “mistake” of any kind (i.e., something that can be avoided).

At first, one might think that this randomness will be eliminated if we increase the precision of measurement. Actually, the opposite is true. For instance, if we measure the length of a table in integer number of feet, the result will be the same under repetition. When we increase the precision, to an inch, then to half of an inch, and so on, the variability of the results under repetition will be increasing.

The question then arises: What inference can be drawn from data which may differ from sample to sample? As a theory, inferential statistics is a part of mathematics, has its own structure of specific concepts (motivated mostly by applications), and its own theorems.

Each such theorem asserts that some conclusions hold, provided that certain assumptions are satisfied. These assumptions refer to the process of data collection, or the properties of underlying random variables.

The applicability of statistical methods depends crucially on the degree to which the assumptions are met in real situations. Sometimes this degree of validity is under the control of the experimenter (see Example 10.1); in some other cases, we simply may have no conclusive evidence that the assumptions are violated; there could also be situations (see Example 10.2) when it may feel justified to disregard the fact that the assumptions are false.

■ EXAMPLE 10.1

Suppose that we want to apply a method of statistical inference, for which we need an assumption that a certain random variable X is binomial. Often X represents the number of elements of some kind in the sample (number of defective items, number of patients who recovered after specific treatment, etc.). The assumption that the process of collecting data is really a sequence of Bernoulli trials depends on various factors, some of which the experimenter can control. Of these, most important is obtaining observations independently. In case of defective items, it may require sampling with replacement; in case of patients, it may require checking that the sample does not have identical twins or other persons whose reaction to the treatment in question may be similar because of some genetic reasons.

■ EXAMPLE 10.2

Imagine that we observe freshmen scores X on an aptitude test. Assuming that X has a normal distribution leads to sufficiently good approximations of the relevant probabilities. The fact remains, however, that test scores cannot be normally distributed, since X must be an integer, and it also cannot assume negative values.

In cases such as above, we typically feel justified in using the theorems that rely on the assumption that X has a normal distribution, even if we know that this assumption is not satisfied. But there may be some “more serious” violations of the assumptions. Checking the validity of an assumption may involve using some elaborate statistical techniques. Often, however, it is sufficient to just have a glance at some preliminary graphical presentation of the data, or at the values of some crude statistics. In either case, we deal with a summary or reduction of the data. The methods of such initial reduction of the data belong to what is called *descriptive statistics*.

A simple summary presentation of the data can lead to the discovery of surprising and important consequences as shown in the three following examples.

The first example below concerns a 1854 cholera outbreak in London during the 1846–1860 worldwide cholera pandemic.

■ EXAMPLE 10.3

Cholera is a deadly disease, its outbreaks were quite common in Europe in the past. The medical community of the nineteenth century London was divided on the cause of the disease. Some doctors were thinking that it was caused by inhaling particles of dirty organic matter in the air, while others were convinced that cholera was caused by the not yet identified germ cell. John Snow, a London physician, had a theory that the germ was transmitted from person to person by drinking contaminated water. He marked cholera incidents on a city map and noticed a cluster of cases near a Broad Street water pump. He then studied water samples from several pumps and found out that it was indeed contaminated water that allowed the spread of the disease rather than particles in the air. In places that switched to cleaner water sources, the disease ceased in few days.

The next two examples both concern the World War II.

■ EXAMPLE 10.4

The main route for supplying the Allied armies fighting Nazi Germany in Europe during World War II was through the Atlantic Ocean. The convoys were attacked regularly, mostly by German U-boats. As the war progressed, more and more data accumulated. It turned out that the average number of ships lost in an attack was relatively constant; in particular, it did not depend on the size of the convoy. This observation led to a simple conclusion: *to decrease losses, make convoys as big as possible*. Indeed, two separate convoys might expect to be detected and attacked about twice as many times as a convoy obtained by combining them. Since the average losses per attack are independent of the convoy size, such joining of two convoys cuts losses by half. This simple idea contributed substantially to winning the war.

■ EXAMPLE 10.5

During the World War II, bombers were sent on missions over Germany. On route to and from, as well as over the target, they were subject to antiaircraft fire. The direct hits were not very frequent, but the AA shells were set to explode at specific altitudes, spraying the planes with shrapnel. The planes that returned from the mission were examined for locations of shrapnel hits, and all these locations were recorded on a silhouette of the plane. As more and more data became available, the silhouette was more densely covered with recorded locations of hits. There were, however, some areas that were hit less often than others.

A surprising order was then given: strengthen (by putting armor plates, etc.) those areas that were hit *seldom*. Here the argument is that the locations of shrapnel hits on bombers must have a uniform distribution over the silhouette of the plane. Any “white areas” in the data therefore indicate the locations of hits for the planes that did not return from missions.

We present these three examples not only because of their combination of simplicity of premises and unexpectedness of conclusion but also because they are all based on elementary ways of representing the data: marking disease incidents on a city grid, plotting average losses per attack against convoy size, or making a scatter diagram of shrapnel hits. Certainly it is not often that one gets a chance of contributing to victory in war or ceasing a pandemic by a visual inspection of some descriptive statistics. Nevertheless, it is worth knowing some simple “tricks of the trade” in presenting the data so as to exhibit certain aspects of interest, or in making certain patterns visible. While graphs are necessary to examine the content and the structure of the data it might also help checking assumptions in statistical models. It is also hard to overestimate importance of the graphs in communicating results of statistical data analysis.

In the first two editions of this text (1997 and 2008), we introduced simple (but often very useful) methods of presenting the data such as dot diagram, box-plot, histogram, and stem-and-leaf diagram often known as a numerical histogram. Today, these graphical methods are usually introduced in introductory statistics courses. However, with the constantly growing data sizes (in both, the number of variables and in the number of records) and more and more powerful computers, new graphical techniques are now becoming available. They can be used to see data in layers, three dimensions, to create statistical maps, to name just a few. One can find these graphical methods in most statistical packages, including R and ggplot2- its library for creating graphics. The R-code for the basic graphical methods can be found in Appendix at the end of the book.

10.2 BASIC MODELS

Statistical methods are used to choose an action (or make decisions) in the situation of partial uncertainty. The uncertainty may be alleviated to some extent by taking observations and identifying factors that affect the consequences of possible actions.

We will introduce two approaches to statistical inference: frequentism and Bayesianism. While they are different and were considered competing for many years, they are becoming more connected in the contemporary statistical methodology.

Now let us imagine ourselves as decision makers. Using the terminology accepted in general decision theory, we say that we are in one of the situations labeled as “world θ_1 ,” . . . , “world θ_n ,” but we do not know which one. It is usually assumed that there is one action appropriate for each “world” (although this assumption is not really necessary). Thus, if we knew the “world” we are in (i.e., knew the value θ_j), we would know which action is the best to take.

The only way to identify the actual “world θ_j ” is to perform some experiments and gather data. Let $\mathbf{X}$ be a generic symbol for the results of such experiments. It may be a single observation of some variable, a sequence of observations, a result of some complicated physical experiment, a score on a battery of tests, and so on, depending on the context. Naturally, to make the entire setup meaningful, there must be some relation between $\mathbf{X}$ and θ_j . This is expressed by the assumption that the observation $\mathbf{X}$, while being random, comes from a distribution that depends on θ_j . The randomness here is the crucial part of the entire setup. It may be due to sampling variability or be inherent in the phenomenon, or both.

It may happen, of course, that observation $\mathbf{X}$ will allow us to identify the value θ_j without ambiguity. This occurs when the sets of possible values of $\mathbf{X}$ are disjoint for different θ_j 's. Such cases, however, are rare in real life. Most often, we face a challenging case when the same outcome $\mathbf{X}$ is possible under several (or even all) θ_j but occurs with different probabilities, depending on θ_j .

■ EXAMPLE 10.6

The examples of situations falling under this scheme abound in science, engineering, management, and in everyday life. A doctor faces a patient who has a headache and high fever. Particular “worlds” are possible illnesses of the patient. The doctor orders some tests, and on the basis of their results $\mathbf{X}$ makes a diagnosis (identifies the illness, perhaps incorrectly) and chooses the best action in view of the illness diagnosed. As another example, take a pharmaceutical company that has developed a drug against a specific disease, hopefully superior to drugs used thus far. The “worlds” may be numbers θ describing the relative advantage of the new drug vs. the best drug available so far, with $\theta > 1$ indicating that the new drug is better, and $\theta \leq 1$ indicating that it is no better (or even inferior) to the drug used thus far. Here, the observation $\mathbf{X}$ is a series of studies, specified in great detail by the U.S. Food and Drug Administration (FDA) standards.

Example 10.6 concerns medicine, but it is obvious that any new method is subject to the same scheme of tests before it is established whether or not (and to what extent) it is superior to some other method.

Statistical theory is concerned with the problem of inference about θ on the basis of $\mathbf{X}$. In the case where there is a choice of the variable that we may observe, the statistical theory is also concerned with choosing the “best” variable (these are called problems of design of experiment). Such statistical theory is known as *classical* or *frequentism*.

But there is also a different approach, known as *Bayesian*. While developed later than the *classical* approach, it is based on a theorem (or a “rule”) named after Reverend Thomas

Bayes who first provided a formula for combining the prior knowledge with the new evidence in his 1763 paper.

Under the Bayesian approach, we assume that it is possible that the user of statistical methods already has some knowledge or previous experience, about the possible “words.” This knowledge makes some of the “words” to appear more likely than others, before any observations $\mathbf{X}$ are taken. Such information can then be updated based on the collected data and used in the process of deciding about θ_j on the basis of observation $\mathbf{x}$ (one can then simply compute the conditional probability of θ_j given $\mathbf{X} = \mathbf{x}$). For example, in the case of a physician seeing a patient with a headache and high fever, the possible “worlds” (in this case illnesses causing this particular set of symptoms) include initial stages of flu as well as the initial stages of the plague (Bubonic plague). The incidence of the latter disease is so rare, however, that the doctor may feel perfectly justified in not ordering any test to check the possibility of the disease being the plague. Here the doctor relies simply on his own and his colleagues’ experience about the incidence of various diseases that might start with fever and headache at a given time and geographic location.

Statisticians who allow probabilities of various “worlds” reflecting the researcher’s experience, intuition, “hunches,” and so on, to be used in statistical methodologies are called *Bayesians*, and the resulting statistical methodology is called *Bayesian statistics* (to be discussed in Chapter 16.4).

A rather strong argument for the Bayesian approach is provided by the theory of outliers (see Section 9.3). In a non-Bayesian setup, where $\theta = (\alpha, \lambda)$ is the parameter in the family of all gamma distributions, no configuration of data values can be rejected as containing an outlier, even if (say) $n - 1$ observations fall into interval $(0, 1)$ and the n th observation exceeds 10^6 . This is because there exists a $\theta = (\alpha, \lambda)$, for which such a configuration of data points is very likely to occur. This counterintuitive example suggests that a statistician should eliminate some domains of the parameter space as “unlikely,” using whatever information or experience he or she has. We will provide Bayesian solutions to various problems under consideration. This means that we will show how the problem is, or may be, solved if the prior probabilities of various θ_j ’s are available. While the basics of the Bayesian statistics will be presented in Chapter 16, a systematic presentation of the theory of Bayesian statistics lies, however, beyond the scope of this book.

10.3 SAMPLING

In the theory of statistics, it is typically assumed that the data are observations of some random variable. The applicability of statistical methods depends therefore on how well the assumption of randomness and the assumption about the distribution are satisfied. In this section, we will show some of the possible “traps” one may encounter in implementing statistical methods in practice.

Let us start with an example.

■ EXAMPLE 10.7

Before the presidential election in 1936, the Literary Digest (one of the most respected magazines of the time) conducted a poll and published a prediction that Alfred Landon would get 57% of the vote against Roosevelt’s 43%. Surprisingly, the actual results of the election were 62% for Roosevelt against 38% for Landon. The irony of the situation was that the Literary Digest poll was one of the largest and most expensive polls ever done, with a sample size of around 2.4 million people! At the same time, George Gallup was able to correctly predict a victory for Roosevelt using a much smaller sample of about 50,000 people.

Unfortunately, poorly designed sampling strategy cannot be improved by increasing the size of the sample. While in general, increasing the size of the sample increases the amount of information and helps to make a better inference, but only if the sample is *representative* of the population. Increasing sample cannot reduce the sampling bias, on the contrary might even inflate it.

The huge Literary Digest's poll bias had two sources, one so-called : *selection bias* and the *nonresponse bias*. Selection bias was caused by the fact that the mailing list was created using telephone directories, lists of magazines' subscribers, and club membership lists—all of which included mostly middle and upper-middle class voters (telephones were the luxury at that time). Nonresponse bias was caused by the low response rate (about 25%). People who respond to surveys are not the same as people who do not.

Suppose that the objective is to estimate an unknown parameter θ that is the average of some attribute in a population. It may be, for instance, the average yearly income of a family in a given region, the average age of a patient receiving certain treatment, and so on. In such cases, one typically takes a sample of elements from the population and measures the values of the attribute being studied. Thus, $\mathbf{X} = (X_1, \dots, X_n)$, where $E(X_i) = \theta$ for $i = 1, \dots, n$.

The following story illustrates some of the potential problems that one may encounter.

■ EXAMPLE 10.8

Three social science students, Jim, Joe, and Susan, were each assigned a task of estimating the average size of a class (number of students) in a given school district. Jim decided to make a card for each class in each school, shuffle the cards, sample one or more of them and then find the number of children in each class sampled.

Joe found a somewhat simpler scheme: He decided to prepare cards with names of schools and first sample a school (or several schools). Then for each school chosen, he decided to make cards with labels of classes and take a sample of those cards, at the end determining the numbers of children in each class sampled.

Susan applied a still simpler scheme: She decided to take a sample of children from the school district and ask each child about the size of the class that he or she attends. The question is: Which of the three students, if any, measured the parameter “average size of the class in a given school district”?

Since an increase of sample size affects only the precision of the estimator, not the parameter that is being estimated, for simplicity we will consider only the cases where Jim, Joe, and Susan each take a single observation.

Suppose that there are k schools in the district in question, with the i th school having n_i classes, of sizes C_{ij} , $i = 1, \dots, k$, $j = 1, \dots, n_i$. Then, the total number of classes is $N = \sum_{i=1}^k n_i$, and the average class size is

$$\theta = \frac{1}{N} \sum_{i=1}^k \sum_{j=1}^{n_i} C_{ij}.$$

The objective is to estimate θ . If X , Y , and Z denote the random variables observed, respectively, by Jim, Joe, and Susan, then it is clear that $X = C_{ij}$ with probability $1/N$ (sampling is from the set of all classes). Thus, $E(X) = \theta$.

For the random variable Y observed by Joe, we have $Y = C_{ij}$ if Joe selects i th school (probability $1/k$) and j th class in the i th school (probability $1/n_i$). Consequently,

$$E(Y) = \sum_{i=1}^k \sum_{j=1}^{n_i} \frac{1}{kn_i} C_{ij} = \frac{1}{k} \sum_{i=1}^k \frac{1}{n_i} \sum_{j=1}^{n_i} C_{ij}.$$

We have $E(Y) \neq \theta$ except in a special case when all n_i 's are equal, meaning each school has the same number of classes.

Finally, for the random variable Z , observed by Susan, the situation is as follows: Let $C = \sum_{i=1}^k \sum_{j=1}^{n_i} C_{ij}$ be the total number of children in all classes. With probability C_{ij}/C , a child from the j th class of the i th school will be selected, and then the value of Z will be C_{ij} . We have therefore

$$E(Z) = \sum_{i=1}^k \sum_{j=1}^{n_i} C_{ij} P\{Z = C_{ij}\} = \sum_{i=1}^k \sum_{j=1}^{n_i} C_{ij} \frac{C_{ij}}{C} = \frac{1}{C} \sum_{i=1}^k \sum_{j=1}^{n_i} C_{ij}^2.$$

Again, $E(Z) \neq \theta$ unless all classes are of the same size.

Thus, it is only Jim whose method provides an estimate of the parameter θ . For random variables suggested by Joe and Susan, we generally have $E(Y) \neq \theta$ and $E(Z) \neq \theta$. The method suggested by Joe is known as *stratified sampling*: The population is divided into *strata*, and one first samples strata, and then takes a sample from each stratum (in this case, the role of strata is played by schools).

It is worth mentioning here that if Joe decided to take a sample of schools, and then collect the data about *all* class sizes in selected schools, he would use what is known as *cluster sampling*.

With some prior information available (e.g., about relative sizes of the strata), one can easily adjust Joe's estimator (by taking appropriate weighted averages) to build random variable Y with $E(Y) = \theta$ (called an *unbiased* estimator of θ).

The situation with the Susan's method is not so straightforward and cannot be easily remedied. It is related to *importance sampling*, where the probability of choosing an element with a larger value of the attribute is higher than the probability of the element with a smaller value.

The bias due to the phenomenon of importance sampling occurs quite often and evades notice. The following example (suggested by R. F. Green, personal communication) provides some surprising insight into the issue.

■ EXAMPLE 10.9 Siblings

Suppose that in a certain society, the distribution of the number of children in a family is Poisson with mean 4. What is the average number of siblings of a child in this society?

SOLUTION. While most intuitive answer seems to be 3, it is actually 4 if we assume Poisson distribution. In general, it is *more* than 3, except when all families have exactly 4 children. The situation is very much the same as with Susan's sampling in Example 10.8.

Before proceeding to the solution, we should note that the distribution given in the problem concerns the population of *families*, but the question concerns the average in the population of *children*. Let $p_0, p_1, \dots$ be the distribution of the number of children in the family, so that in the special case under analysis, we have $p_k = (\lambda^k/k!)e^{-\lambda}$ for $\lambda = 4$. Then, $\mu = p_1 + 2p_2 + \dots$ and $\sigma^2 = \sum k^2 p_k - \mu^2$ are the mean and variance of the distribution $\{p_k\}$. Suppose that the population consists of a large number N of families. Clearly, Np_0 families have no children, Np_1 families have one child, and so on. The total size of the population of children is therefore

$$M = Np_1 + 2Np_2 + 3Np_3 + \dots = N\mu.$$

The probability of choosing a child from a family with k children is

$$\frac{kNp_k}{N\mu} = \frac{kp_k}{\mu}$$

for $k = 1, 2, \dots$ and such a child has $k - 1$ siblings. Then, the average number of siblings is

$$\begin{aligned} Q &= \sum_{k=1}^{\infty} (k-1) \frac{kp_k}{\mu} = \frac{1}{\mu} \left\{ \sum_{k=1}^{\infty} k^2 p_k - \sum_{k=1}^{\infty} kp_k \right\} \\ &= \frac{1}{\mu} \{\sigma^2 + \mu^2 - \mu\} = \frac{\sigma^2}{\mu} + \mu - 1. \end{aligned}$$

As can be seen, we have $Q \geq \mu - 1$ with $Q = \mu - 1$ in the case where $\sigma^2 = 0$ (i.e., when all families have the same number μ of children). For the Poisson distribution $\sigma^2 = \mu = \lambda$, so we have $Q = \mu = 4$.

A sampling bias closely related to the bias from importance sampling is connected with the following phenomenon, which caused some controversy before it became properly understood. To explain it, we will again use an anecdotal example.

■ EXAMPLE 10.10 Renewal Paradox

An objective is to estimate the average lifetime of electric bulbs of a particular type, all produced by the same company. We assume that the distribution of the lifetime T is exponential with mean $E(T) = 1/\lambda$.

The usual procedure is to take a random sample of bulbs and observe their lifetimes $T_1, T_2, \dots, T_N$. Such data can be used to estimate $E(T)$. If testing is run in parallel, it takes time $T_{N:N} = \max\{T_1, \dots, T_N\}$ to collect the data. One could speed up the procedure by observing only the k shortest lifetimes $T_{1:N} < T_{2:N} < \dots < T_{k:N}$ and then interrupt the data collection, recording only that $T_{k+1:N} > T^*$, where T^* is some threshold (so that $N - k$ lifetimes are not fully observed, but they are all known to exceed T^*). This is called *censoring of the data*.

In some cases, another way can be used to collect data, that does not involve waiting. Suppose that there is a large building where the maintenance personnel use only bulbs of the type of interest for us. Whenever a bulb fails, it is immediately replaced by a new bulb, and this change is recorded. As a consequence, we have access to the records, reaching into the past, of the time of replacement of every bulb. We might then select some time t^* (preferably in the past) and use all lifetimes of the bulbs that were operating at time t^* as the sample. Clearly, if t^* is sufficiently far back in the past, each of the bulbs that operated at t^* has already been replaced, and their lifetime $T(t^*)$ is known. Otherwise, we would have to wait for some time to observe the value of $T(t^*)$ for that bulb.

Let T'_i be the *spent* lifetime at t^* for the i th bulb, and similarly, let T''_i be the *residual* lifetime at t^* . Moreover, let t_p denote the current time. It is clear that if $t_p - t^*$ is large enough, all residual times will be observable and their lifetimes will be $\bar{T}_i = T'_i + T''_i$, otherwise residuals will be at least $t_p - t^*$ and consequently lifetimes will be at least $T'_i + t_p - t^*$. The sample $T_1, T_2, \dots, T_N$ of all N bulbs that were operating at t^* can be observed and the average $(1/N) \sum_{i=1}^N T_i(t^*)$ can serve as an estimator $E(T)$.

The obvious question is: Is such a method of collecting data correct? It should be stressed that we are *not* concerned here with the practical implementation of the scheme. We assume that the time t^* is chosen without the knowledge of the replacement records (a condition preventing conscious or unconscious bias) and disregard the fact

that in reality light bulbs are used only part of the time, and that at some locations lights are used more often or are subject to different conditions (outdoor and indoor lights, etc.). Since the problem concerns theory, we deliberately idealize the situation and assume that the data concern only bulbs that operate constantly and under the same conditions. This way the interreplacement times are sampled independently from the same exponential distribution.

To find the answer, consider a single process of changing of bulbs in one place. The process starts at time $t = 0$; the consecutive lifetimes are $\xi_1, \xi_2, \dots$, and the times of replacements are $S_0 = 0, S_n = S_{n-1} + \xi_n$ for $n \geq 1$. The lifetime recorded in the sample, $T(t^*)$, is the value $\xi_n = S_n - S_{n-1}$ such that $S_{n-1} \leq t^* < S_n$. We assume that $\xi_1, \xi_2, \dots$ are iid with exponential distribution.

The distribution of the time S_n of the n th replacement was derived in Section 8.3. It is a gamma distribution with parameters n and λ so that the density of S_n is

$$g_n(t) = \frac{\lambda^n}{\Gamma(n)} t^{n-1} e^{-\lambda t} = \frac{\lambda^n}{(n-1)!} t^{n-1} e^{-\lambda t}. \quad (10.1)$$

For further use, observe that we have

$$\sum_{n=1}^{\infty} g_n(t) = \lambda \sum_{n=1}^{\infty} \frac{(\lambda t)^{n-1}}{(n-1)!} e^{-\lambda t} = \lambda. \quad (10.2)$$

To find $E[T(t^*)]$, we must first find $F_{T(t^*)}(x) = P\{T(t^*) \leq x\}$, the cdf of $T(t^*)$. Assume that $x < t^*$. Then a replacement would have occurred before time t^* , since otherwise the original bulb would still be working at t^* , and we would have $T(t^*) = \xi_1 > t^* > x$.

Thus, for some $n = 2, 3, \dots$ we must have $S_{n-1} = z \leq t^* < S_n = S_{n-1} + \xi_n$. In this case, $\xi_n > t^* - z$ so that $t^* - z < \xi_n \leq x$. The condition $\xi_n \leq x$ implies $t^* - z \leq x$, which means that the time z of the $(n-1)$ st replacement satisfies $t^* - x \leq z \leq t^*$. Partitioning with respect to $n = 2, 3, \dots$, conditioning on time z (time of the $(n-1)$ st replacement), and using (10.2), we obtain

$$\begin{aligned} F_{T(t^*)}(x) &= P\{T(t^*) \leq x\} = \sum_{n=2}^{\infty} \int_{t^*-x}^{t^*} P\{t^* - z < \xi_n \leq x\} g_{n-1}(z) dz \\ &= \sum_{n=2}^{\infty} \int_{t^*-x}^{t^*} [e^{-\lambda(t^*-z)} - e^{-\lambda x}] g_{n-1}(z) dz \\ &= \int_{t^*-x}^{t^*} [e^{-\lambda(t^*-z)} - e^{-\lambda x}] \sum_{n=2}^{\infty} g_{n-1}(z) dz \\ &= \lambda \int_{t^*-x}^{t^*} [e^{-\lambda(t^*-z)} - e^{-\lambda x}] dz = 1 - e^{-\lambda x} - \lambda x e^{-\lambda x}. \end{aligned}$$

If $x > t^*$, the derivation above has to be modified in two ways. First, we have to add the probability that the first bulb is still working at t^* , and that its lifetime ξ_1 satisfies $t^* < \xi_1 \leq x$ (see Figure 10.1). Second, if a replacement occurred before t^* , the time

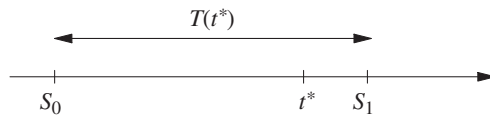


Figure 10.1 Lifetime $T(t^*)$ of a bulb.

$S_{n-1} = z$ of last replacement before t^* satisfies the inequality $0 < z \leq t^*$. Consequently, we have

$$\begin{aligned} P\{T(t^*) \leq x\} &= e^{-\lambda t^*} - e^{-\lambda x} + \sum_{n=2}^{\infty} \int_0^{t^*} [e^{-\lambda(t^*-z)} - e^{-\lambda x}] g_{n-1}(z) dz \\ &= 1 - e^{-\lambda x} - \lambda t^* e^{-\lambda x}. \end{aligned}$$

Differentiating, we obtain the density of $T(t^*)$:

$$f_{T(t^*)}(x) = \begin{cases} \lambda^2 x e^{-\lambda x} & \text{for } x < t^* \\ \lambda(1 + \lambda t^*) e^{-\lambda x} & \text{for } x \geq t^*. \end{cases} \quad (10.3)$$

An easy integration gives the formula for $E[T(t^*)]$, and we can show that

$$\lim_{t^* \rightarrow \infty} E[T(t^*)] = \frac{2}{\lambda}.$$

Therefore, on average, the lifetime of the bulb that is in operation at time t^* is about *twice as long* as the average lifetime of other bulbs. So the described method of sampling is biased.

The conclusion above about sampling procedure being biased remains valid as long as the lifetimes $\xi_1, \xi_2, \dots$ are iid. The specific assumption of exponentiality of distribution implies the form (10.3) of the density, but the fact that $\lim_{t^* \rightarrow \infty} E[T(t^*)] > E(\xi)$ is valid for any nondegenerate distribution of ξ_i 's (e.g., see Karlin and Taylor, 1975).

Since $T(t^*) = T' + T''$, meaning $T(t^*)$ is the spent lifetime at t^* plus the residual lifetime at t^* , we can try to derive our result as follows: The residual time T'' is exponential, in view of the memoryless property of exponential distribution, so $E(T'') = 1/\lambda$. For the spent lifetime T' , it cannot exceed t^* , and it exceeds x (where $x < t^*$) if there are no replacements between $t^* - x$ and t^* . Since replacements form a Poisson process, the latter probability is $e^{-\lambda(t^*-x)}$. Consequently

$$P\{T' > x\} = \begin{cases} e^{-\lambda(t^*-x)} & \text{for } x < t^* \\ 0 & \text{for } x \geq t^*. \end{cases}$$

Thus, by Theorem 7.2.2

$$E(T') = \int_0^{\infty} P\{T' > x\} dx = \int_0^{t^*} e^{-\lambda(t^*-x)} dx = \frac{1}{\lambda}(1 - e^{-\lambda t^*}),$$

and therefore $E[T(t^*)] = E(T') + E(T'') = 2/\lambda - (1/\lambda)e^{-\lambda t^*}$. This calculation, however, relies on the assumption that the memoryless property of the exponential distribution is valid for residual waiting time counted from a *randomly selected* moment (and not only from a fixed moment).

For example, suppose that we observe the waiting time for nearest replacement from the moment *preceding* a given replacement by some fixed constant, say 5 hours. Such a moment is random, and yet the waiting time is at most 5 hours by definition, and therefore does not have exponential distribution. This simple example shows that the notion of a “random moment” has to be qualified: We are not allowed to know the future at the time we select our moment to start the observations.

10.4 MEASUREMENT SCALES

Data sets can be of different types, have different characteristics, and different types require different statistical methods. We will now introduce some concepts related to the level of measurements. Consider a set of objects of some kind. Typically, for the purpose of analysis, statistical description, and so on, each of these objects can be represented by a number $b(x)$ assigned to object x . Often that number represents the result of measurement in some units, but sometimes it identifies only the class to which the object belongs.

If $b(x) > b(y)$, then x has “more . . .” than y (is heavier, longer, warmer, older, etc.). The question we want to address is which type of statements expressed through the values of b are meaningful and which are not. To take an example, if b represents length, and $b(x) = 10$, $b(y) = 5$, we say that x is *twice as long* as y , since $b(x)/b(y) = 2$. This statement will remain valid whether we express the length in inches, centimeters, or miles: the values $b(x)$ and $b(y)$ will change, but their ratio will remain the same. However, if b represents temperature and $b(x) = 10$, $b(y) = 5$, it makes no sense to say that x is “twice as warm” as y . Indeed, it is enough to express temperature on a different scale (e.g., change from Fahrenheit to Celsius): The ratio of temperatures will change as the scale changes. The question therefore is: Why are the ratios of scale values meaningful for length, duration, or weight, but not meaningful¹⁹ for temperature?

The full impact of such questions became apparent only when physical, mathematical, and statistical methods started to be used in the social sciences. Here the attributes that one considers are typically “soft,” and the use of certain methods can lead to conclusions that are illegitimate, but have the deceptive appearance of being very precise (the statement that a new brand of instant coffee “tastes 11.3% better” may be effective in commercials, but its meaning, if any, is obscure).

It is clear that the decision which statements expressed through the values $b(x)$ are allowed and which are not must lie in the analysis of the nature of a measured attribute.

The measurement is an assignment of numbers to objects in some set A (hence it is a function on A). This function must satisfy conditions that “mimic” the empirical relations $R_1, R_2, \dots$. If such a function exists, we say that measurement exists.

The following examples illustrate the above description.

■ EXAMPLE 10.11 Scale of Hardness

Let the objects under consideration be minerals, and let the empirical relation $\succ$ be defined as $x \succ y$ if mineral x scratches mineral y . We will assign numbers $h(x)$ to objects x in A (minerals) to represent their hardness, in the sense that $h(x) > h(y)$ whenever $x \succ y$. One such assignment gives the value 10 to diamonds and lower values to other minerals. This choice of assignment of numbers to minerals is arbitrary, except that the relation $\succ$ between values of function h must mimic the relation $\succ$ between arguments of h (i.e., minerals). It is precisely this degree of arbitrariness of h that makes it meaningless to claim that “ x is twice as hard as y ” if $h(x)/h(y) = 2$.

But even in such a simple case as above, the possibility of assigning numerical scale values $h(x)$ to minerals x in A results from the fact that the relation $\succ$ (of scratching) is transitive: if $x \succ y$ and $y \succ z$, then $x \succ z$. Still the existence of a function h that

¹⁹Observe that the precision of measurements has nothing to do with the answer. Ratios of lengths make sense even if the lengths are determined imprecisely; the ratios then are simply subject to bigger errors. But the ratios of the temperatures make no sense even if the temperature is measured with the most precise devices available.

mimics the relation $\succ$ is not obvious. Imagine that infinitely many new minerals, each of different hardness, are suddenly discovered. Could one still find enough distinct numbers to label those minerals? The answer is positive, but we will not provide the proof here.

■ EXAMPLE 10.12

Consider now an attribute such as length and a possible measurement system for it. We have here the relation of comparisons with respect to length, accomplished empirically by putting the objects parallel to each other, with one end lined up (like pencils). We then obtain a relation, $\succeq$, interpreted as “not shorter than.” The ultimate goal is to show the existence of a function b (length) defined on A , such that $x \succeq y$ if, and only if, $b(x) \geq b(y)$. Clearly, we must require that $\succeq$ satisfies some conditions (axioms), such as transitivity: if $x \succeq y$ and $y \succeq z$, then $x \succeq z$, and so on.

However, one relation $\succeq$ is not sufficient to define length as we know it, since there may be many functions b satisfying the requirement that $\succeq$ agrees with the order induced by the values of b . Clearly, something more is needed to force all functions b to differ one from another only by the unit of measurement (i.e., such that if b and b^* are two assignments of lengths, then $b^* = \alpha b$ for some $\alpha > 0$).

In this case we need a ternary relation describing the operation of concatenation, that is, putting the objects end to end. Letting $\circ$ denote such an operation, we obtain new objects, such as $x \circ y$, corresponding to x and y aligned one after another. This operation can be identified with a relation that holds between x, y , and z , if $x \circ y \sim z$, where $\sim$ is defined by $a \sim b$ if $a \succeq b$ and $b \succeq a$. Clearly, we want $b(x \circ y) = b(x) + b(y)$, and to achieve that, the relations $\succeq$ and $\circ$ jointly must satisfy a number of conditions, such as the most obvious ones: if $x \succeq y$, then $x \circ z \succeq y \circ z$ for every z , and $(x \circ y) \circ z \sim (z \circ x) \circ y$, for all x, y, z . Less obvious is the requirement of the Archimedean property:

For every x and y there exists n such that $x^{\circ n} \succeq y$,

where $x^{\circ n} = x \circ \dots \circ x$ (n times).

A measurement theory of length is then a relational system consisting of set A , relations $\succeq$ and $\circ$, a set of conditions for $\succeq$ and $\circ$ (referred to as *axioms*), and a theorem asserting that if these axioms are satisfied, then:

1. *There exists a real-valued strictly positive function b on A such that*

$$b(x \circ y) = b(x) + b(y) \text{ and } x \succeq y \text{ if and only if } b(x) \geq b(y).$$

2. *If b^* is any other function satisfying condition 1, then $b^* = \alpha b$ for some $\alpha > 0$.*

The first part of the assertion provides the existence of measurement b , and the second part provides information about its uniqueness. In the case of length, the measurement is unique up to the choice of unit of measurement, so that the ratios of lengths are invariant. This means that for any objects x, y the ratio $b(x)/b(y)$ does not depend on the choice of function b .

Each scale has its own level of “uniqueness,” described by a condition corresponding to condition (2) in Example 10.12. Although theoretically there may be infinitely many types of scales, in practical situations one encounters only four major types of scales, as specified in Definition 10.4.1.

Consider a relational system, where b and b^* are any two assignments of numbers to objects in the set A —measurement scales specified by the axioms of the relational system.

Definition 10.4.1

- (i) If there exists $\alpha > 0$ such that $b^*(x) = \alpha b(x)$, then the measurement is on the *ratio* scale.
- (ii) If there exists $\alpha > 0$ and β such that $b^*(x) = \alpha b(x) + \beta$, then the measurement is on the *interval* scale.
- (iii) If there exists a monotone increasing function u such that $b^*(x) = u[b(x)]$, then the measurement is on the *ordinal* scale
- (iv) If there exists a one-to-one function v such that $b^*(x) = v[b(x)]$, then the measurement is on the *nominal* scale. □

In other words, for ratio scales (see Example 10.12), the measurement is unique up to a choice of unit (examples are length, duration, etc.). For interval scales, one can choose not only the unit but also the zero of the scale (e.g., temperature). For ordinal scales only the order matters. For instance, if $b(x) = 10$ and $b(y) = 2$, then all that we can say is that x is “more . . .” than y , on an attribute designated by We cannot meaningfully say that the “difference” between x and y is 8, or that x is “five times . . .” as y . The reason is that $b(x) = 10$ and $b(y) = 2$ can be replaced by $b^*(x) = 100$, $b^*(y) = -3$, or $b^*(x) = 1$, $b^*(y) = 0.99$, or any other two values, as long as the first exceeds the second.

In physical measurement, the best such scale known is that of hardness (see Example 10.11) when the main empirical relation is $x \succeq y$ if “ x scratches y .” In the social sciences, the situation is not so clear. The relations between stimuli elicited by asking subjects to evaluate them “on the scale of 1 to 10” or by marking responses such as “strongly agree,” “agree,” and so on, are of the ordinal nature but often are incorrectly treated as if they are expressed on an interval or ratio scale.

The same concerns education, where grades are averaged and compared as if they were measured on an interval scale. Thus a student who takes six classes and gets one *A* and five *B*’s as the semester grades has a GPA of 3.16. Another student, who also takes six courses, gets four *A*’s, and two *C*’s, has a GPA that is higher (3.33). If the scoring (function b defined on grades *A*, *B*, *C*, etc.) were changed to $b(A) = 4$, $b(B) = 3$, $b(C) = 1$, that student’s GPA would be 3.00 and his grades would be judged as worse than these of the first student.

There is nothing unique or objective in assigning the values 4, 3, and 2 to grades *A*, *B*, and *C*, considering that the process of averaging grades usually includes different subjects, grades by different teachers, and with criteria often formulated rather vaguely. Nevertheless, tradition and the practical need to assess students’ performance force one to fix the scoring system for grades, and regard them as a measurement on an interval scale.

Finally, the nominal scale—the weakest of the four—can hardly be regarded as measurement, since the numbers serve only as labels, for the sole purpose of identification (e.g., numbers on jerseys of football players).

The four scales mentioned above form an order: any transformation $b^*(x) = \alpha b(x)$, $\alpha > 0$ allowed for the ratio scale is a particular case of a transformation $b^*(x) = \alpha b(x) + \beta$, $\alpha > 0$ that defines the interval scale. This transformation is monotone, hence allowed for the ordinal scale, and in turn, each monotone transformation is one to one, allowed for the nominal scale. One can always lower the level of measurement, while to use a higher measurement level is incorrect.

The theory of measurement scales, as outlined above, was introduced by Stevens (1946). It rapidly gained popularity in psychology and the social sciences and was developed into a highly sophisticated theory (e.g., see Krantz et al., 1971, or Roberts, 1979). The four types of scales, as defined above, are most commonly encountered and popularly known.

The types of statistical methods that one can sensibly use depend on the type of the scale. While the mean μ and standard deviation σ are defined in both cases, one should not use the coefficient of variation σ/μ in the case of data values measured on the interval (but not ratio) scale. This is because under transformation $y = \alpha x + \beta$ ($\alpha > 0$) the standard deviation

becomes multiplied by α , while the mean becomes multiplied by α and shifted by β . Therefore the value σ/μ depends on the choice of zero of the scale.

On the other hand, for data measured on the ordinal scale only, the mean and standard deviation are not invariant, and only statistics expressed through ranks (e.g., median and other quantiles) should be used. Finally, for data on the nominal scale, one is allowed to use only class frequencies.

In Chapters 11–13, it is assumed that the data represent measurements on the interval scale. This allows us to use such characteristics as the mean and standard deviation, and to assume that the data follow a normal distribution.²⁰ Chapter 14 concerns methods of handling the data measured on ordinal scales only (more precisely, only when ordinal relations are taken into account). Finally, in Chapter 15 we discuss the case of *categorical* data, expressed on the nominal and/or ordinal scale with data grouped into classes.

²⁰Strictly speaking, if X is measured on a ratio scale, it cannot have normal distribution (since negative values are ruled out for ratio scales). Such level of adherence to the rules, imposed by scale types, would drastically impoverish the scope of statistical applications by absurdly disallowing to treat attributes such as height, weight, size, duration, etc., in certain populations as normally distributed, merely because these attributes cannot be represented by a negative number.